

Diagnostic Accuracy of 3D Deep Learning Classifiers for Glaucoma Detection: A Comparison of Cross-Domain and Device-Specific Models

Akram Belghith¹, Christopher Bowd¹, Robert N. Weinreb¹, Jalil Jalili¹, Mark Christopher¹, and Linda M. Zangwill¹

¹ Hamilton Glaucoma Center and Division of Ophthalmology Informatics and Data Science, Shiley Eye Institute, The Viterbi Family Department of Ophthalmology, UC San Diego, La Jolla, CA, USA

Correspondence: Linda M. Zangwill, 9500 Gilman Drive, La Jolla, CA 92093-0946, Shiley Eye Institute/Hamilton Glaucoma Center, Viterbi Family Department of Ophthalmology, University of California, San Diego, CA 92093, USA. e-mail: lzangwill@health.ucsd.edu

Received: April 4, 2025

Accepted: July 12, 2025

Published: August 20, 2025

Keywords: glaucoma; cross-domain; optical coherence tomography; deep learning; classification

Citation: Belghith A, Bowd C, Weinreb RN, Jalili J, Christopher M, Zangwill LM. Diagnostic accuracy of 3D deep learning classifiers for glaucoma detection: A comparison of cross-domain and device-specific models. *Transl Vis Sci Technol*. 2025;14(8):29. <https://doi.org/10.1167/tvst.14.8.29>

Purpose: To evaluate the accuracy of a three-dimensional (3D) deep learning (3D DL) and 3D cross domain deep learning (3D CD-DL) classifiers compared to standard macular ganglion cell-inner plexiform layer (GCIPL) thickness measurements for classifying eyes with glaucoma using optical coherence tomography (OCT).

Methods: A total of 502 primary open-angle glaucoma eyes from 295 patients and 119 healthy eyes from 63 individuals were included. Two classifiers were compared: (1) a 3D DL model trained on Spectralis macular OCT and applied to Spectralis macular OCT images and (2) 3D CD-DL model trained on synthetic Spectralis images generated from 3D Cirrus macular OCT using Cycle-consistent adversarial networks (CycleGAN) and applied to real Spectralis macula OCT images. An additional 100 different eyes (50 Cirrus, 50 Spectralis) were used to train the CycleGAN. Age, axial length, and disc area adjusted area under the receiver operating curves (AUROC) were used to compare model accuracy.

Results: Adjusted AUROC for 3D DL model was 0.92 (95% confidence interval [CI], 0.85–0.95). This was significantly higher than global GCIPL thickness (0.83 [0.78–0.85], $p \leq 0.001$) but similar to 3D CD-DL (0.91 [0.84–0.95], $P = 0.45$). Using only early glaucoma eyes (mean deviation ≥ -3.0 dB), the 3D DL model showed significantly higher diagnostic accuracy (0.90 [0.84–0.94]) compared to global GCIPL thickness (0.80 [0.76–0.82], $P \leq 0.001$) but similar to the 3D CD-DL model (0.90 [0.83–0.93], $P = 0.51$).

Conclusions: The 3D DL classifier showed significantly higher diagnostic accuracy than global GCIPL thickness but was similar in performance to the 3D CD-DL classifier. By using synthetic data and diverse training sets, cross-domain learning produces robust, generalizable models across different imaging devices as demonstrated by the comparable accuracy of the 3D CD-DL and device-specific 3D DL models. More data from other OCT devices are needed to further validate these findings.

Translational Relevance: The 3D Deep learning models significantly surpass traditional GCIPL thickness measurements for accurately detecting glaucoma. The cross-domain model closely matches the performance of the device-specific model in glaucoma classification potentially reducing the need for device-specific models in clinical practice.

Introduction

Despite advancements in diagnostics and treatment over the past 20 years, glaucoma remains a leading cause of irreversible blindness.^{1,2} In 2020, approxi-

mately 76 million people were living with glaucoma, and this number is projected to exceed 111 million by 2040.^{3,4} The condition involves the progressive loss of retinal ganglion cells (RGCs) and their axons, leading to optic neuropathy and visual field loss, ultimately causing permanent blindness.⁵

Deep learning (DL) has significantly impacted the field of ophthalmology, offering advanced tools for the diagnosis and management of glaucoma largely through medical image analysis. For instance, our group⁶ developed a two-dimensional (2D) convolutional neural network (2D-CNN) approach that uses wide-angle swept-source optical coherence tomography (OCT) images to extract features from the retinal nerve fiber layer (RNFL) to improve glaucoma classification. This technique outperformed traditional measures such as circumpapillary RNFL thickness, standard automated perimetry mean deviation (MD) and frequency doubling perimetry MD. More recently, Mehta et al.⁷ introduced a multimodal framework that integrates various ocular and clinical data for glaucoma detection. This model used a 2D-CNN structure based on Densenet201 to analyze individual B-scans from OCT volumes obtained from the UK Biobank. By synthesizing data from multiple OCT B-scans, the model achieved an area under the receiver operating characteristic curve (AUROC) of 0.950. Additionally, we recently proposed a multimodal deep learning classifier⁸ using wide-field OCT optic nerve head (ONH) images to classify primary open-angle glaucoma (POAG) and healthy eyes. The combined model outperformed individual models for texture-based en face images and for RNFL thickness maps. Moreover, this method was effective for high myopic eyes.

Despite these advancements, a key challenge remains: domain shift, where models are often limited to images acquired from the same device on which they were trained, reducing generalizability across platforms. Maetschke et al.⁹ proposed a 3D-CNN algorithm for glaucoma detection which directly classified healthy and glaucomatous eyes using unsegmented OCT volumes from the ONH region with favorable results. Because this method did not rely on proprietary segmentation algorithms, it held promise for the potential analysis across measurements from different OCT instruments.

In a recent attempt to address cross-platform analysis issues (i.e., analyzing measurements from different imaging instruments), Wu et al.¹⁰ proposed an unsupervised cross-domain segmentation algorithm using generative adversarial networks (GANs) to identify intraretinal fluid and retinal layers in macular OCT images from different manufacturers. Their algorithm, GANSeg, segmented images from the Topcon 1000 OCT using labeled data from Heidelberg Spectralis OCT images. GANSeg achieved Dice scores comparable to those of experts, with the highest performance in segmenting the ganglion cell-inner plexiform layer (90%; 95% confidence interval [CI], 68%–96%).

The aim of the current study was to use synthetic images to assess the diagnostic accuracy of a 3D cross-domain deep learning (3D CD-DL) classifier in comparison to a standard 3D deep learning (3D DL) classifier and conventional measurements of macular ganglion cell-inner plexiform layer (GC IPL) thickness to detect glaucoma. We hypothesized that by using synthetic data designed to replicate the characteristics of each imaging device, the 3D CD-DL model could effectively address the domain shift issue, providing a more generalizable tool for glaucoma detection across various OCT platforms.

Material and Methods

This study was carried out in accordance with the principles of the Declaration of Helsinki. The University of California San Diego Institutional Review Board and Human Subjects Committee approved all protocols and methods. Participants included in this cross-sectional observational study were recruited from the Diagnostic Innovations in Glaucoma Study (DIGS) (registered at <http://clinicaltrials.gov> (no. NCT00221923) on September 14, 2005). Details of the DIGS protocol have been described previously.¹¹

Participants

Available data included demographics, ocular and medical history, best-corrected visual acuity, slit-lamp biomicroscopy, intraocular pressure (IOP) measured by Goldmann applanation tonometry, gonioscopy, central corneal thickness measured by ultrasound pachymetry, mydriatic fundus examination, simultaneous stereophotography of the optic disc, visual field (VF) testing using standard automated perimetry (Humphrey Field Analyzer; 24-2 with Swedish Interactive Thresholding Algorithm), and Cirrus and Spectralis OCT imaging.

Inclusion criteria were age ≥ 18 years, open anterior chamber angles on gonioscopy, and best-corrected visual acuity of 20/40 or better at baseline. Patients with ocular or systemic diseases affecting the VF or optic nerve, including peripapillary staphylomas, were excluded. Unreliable VFs and poor-quality OCT scans were also excluded.

All VFs were evaluated by UC San Diego Visual Field Assessment Center using a standardized protocol. VF results with fixation losses or false-positive errors over 33% and those classified as “abnormally high sensitivity” by the Glaucoma Hemifield Test (GHT) were excluded. GHT is an algorithm used in

automated perimetry to assess asymmetry in the visual field between the superior and inferior hemifields, categorizing results as within normal limits, borderline, or outside normal limits

POAG was defined by glaucomatous VF loss (pattern standard deviation < 95% confidence limits or GHT outside normal limits) with corresponding ONH damage confirmed via stereophotograph assessment. This dual criterion was used to minimize the risk of misclassification caused by non-glaucomatous VF defects. Stereo photograph-based glaucomatous damage was based on the presence of focal or diffuse neuroretinal rim narrowing or RNFL defects indicative of glaucoma, assessed by two masked graders. In cases of disagreement, a senior grader made the final diagnosis. Healthy individuals had IOP $\leq$ 21 mm Hg, no history of elevated IOP, normal optic discs, intact neuroretinal rims and RNFLs, and at least two reliable and normal VFs with pattern standard deviation within 95% confidence limits with GHT results within normal-limits OU.

OCT

Each subject was required to have high-quality Cirrus and Spectralis SD-OCT scans taken on the same day to ensure a direct comparison of the same eye under identical physiological conditions. Same-day acquisition also mitigated inconsistencies caused by factors such as hydration and measurement conditions, ensuring the reliability and validity of the model comparison. Cirrus HD-OCT 4000 (Carl Zeiss Meditec, Dublin, CA, USA) fovea-centred macular scans were acquired and processed using software version 6.5.0.772. The Cirrus HD-OCT has an acquisition rate of 27,000 axial scans (A-scans) per second with an axial resolution of 5 μ m. The instrument's 840-nm wavelength laser beam generates a cube of data measuring 6 $\times$ 6 $\times$ 2 mm after scanning a series of 200 B-scans with 200 A-scans per B-scan with each A-scan consisting of 1024 voxels. Images in DICOM format, were electronically transferred to the University of California San Diego Imaging Data Evaluation and Assessment (IDEA) Center for quality assessment. Quality review by IDEA Center personnel required well-centred scans with signal strength $\geq$ 7 dB in the absence of movement artifacts.

Spectralis SD-OCT system (Heidelberg Engineering, Heidelberg, Germany) fovea-centred macula scans were analysed using software version 6.0.9.0. Spectralis has an acquisition rate of 40,000 A-scans per second and provides an axial resolution of 3.9 μ m. The macular scan, covering a 4.5 $\times$ 4.5 mm area, is composed of 73 B-scans at a 30° field of view, with each

B-scan containing 768 A-scans. Images in .vol format were electronically transferred to the IDEA Center for quality assessment. Quality review of Spectralis images by IDEA Center personnel required well-centered scans with signal strength >15 dB in the absence of movement artifacts.

GCIPL Thickness Map Extraction

Raw 3D macula SD-OCT volume scans were exported to a mathematical computing software (MATLAB; MathWorks, Natick, MA, USA). The San Diego Automated Layer Segmentation Algorithm-deep¹² was used to automatically segment the RNFL and the inner plexiform layer, enabling the calculation of the macular GCIPL thickness map. Here, we used the average global values from the 1 mm and 3 mm diameter circles of the grid, excluding the central 1 mm diameter area corresponding to the foveola. This exclusion was necessary because of the anatomical variability of the foveal pit, which may not be reliable for glaucoma detection. These values were used directly in the analysis to discriminate between healthy and glaucomatous eyes, based solely on the global thickness value for each eye and no machine learning classification model was applied to GCIPL thickness. Further methodological detail and its application to ophthalmology is available in Medeiros et al.¹³

Cross Domain Convolutional Neural Network

GANs use deep learning techniques, such as convolutional neural networks, to generate realistic synthetic data by identifying patterns in real images. The model consists of two sub-networks trained adversarially: (1) a generator, which creates synthetic images, and (2) a discriminator, which learns to distinguish between real and generated images, improving the generator's output over successive training iterations. These two models are trained together adversarially until the generator produces realistic examples that the discriminator cannot reliably distinguish from real images. Cycle-consistent adversarial networks (CycleGAN) were recently developed¹⁴ to generate images without paired datasets.

CycleGAN trains two generators and two discriminators simultaneously, enforcing a cycle-consistency constraint to ensure that an image translated from one domain to another and back remains similar to the original. This architecture enables translation between image domains without the need for paired data. Because paired Cirrus-Spectralis scans were not available, CycleGAN was an appropriate choice for this task. In our study, we used it to generate synthetic

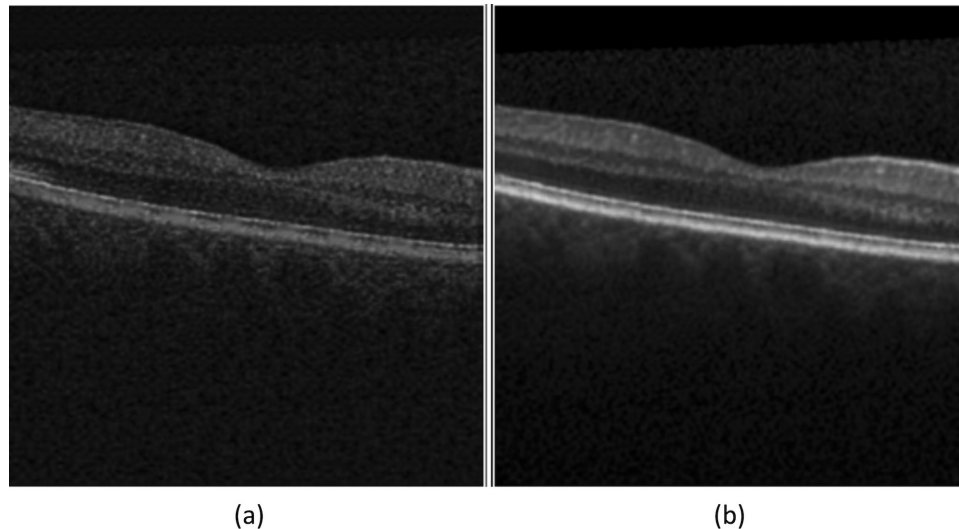


Figure 1. (a) A real Cirrus B-scan from the Cirrus instrument and (b) the corresponding synthetic B-scan mimicking the Spectralis instrument.

Spectralis-like B-scans from Cirrus images for use in training the classifier. To ensure spatial and anatomical consistency between Cirrus and Spectralis volumes, we performed a series of preprocessing steps on the Cirrus macular cube scans ($200 \times 200 \times 1024$ over $6 \times 6 \times 2$ mm). First, we cropped a central 4.5×4.5 mm fovea-centered region to match the field of view of Spectralis volumes, resulting in 150×150 A-scans and B-scans. We then selected 73 evenly spaced B-scans from this subset to align with the Spectralis y-axis layout. To match the higher lateral resolution of Spectralis, each B-scan was resampled from 150 to 768 A-scans. Along the axial (z) dimension, Cirrus volumes originally covered 2.0 mm with 1024 samples. To align with Spectralis, which captures approximately 1.9 mm in depth with 496 slices, we cropped the Cirrus depth to the first 973 slices (corresponding to 1.9 mm) and then resampled to 496 slices using trilinear interpolation. This resulted in final preprocessed Cirrus volumes of size $768 \times 73 \times 496$, matching the Spectralis dimensions used for CycleGAN training and 3D model input.

The model was trained on Cirrus and Spectralis OCT scans from different patients, ensuring no overlap between training and validation sets. Specifically, the training set included 50 Cirrus and 50 Spectralis macular volume scans (40 glaucomatous, 10 healthy per device), all from DIGS participants who were not included in the study cohort used for evaluation. The glaucomatous eyes exhibited a range of disease severity, from early to moderate glaucoma, with a mean deviation (MD) value of -4.7 dB (95% CI, -3.5 to -5.8), ensuring model exposure to a diverse spectrum of glaucomatous damage.

In total, the model was trained on 7300 B-scans (3650 per domain). Prior CycleGAN studies, such as Zhu et al.,¹⁴ successfully trained models with only 1000 images per domain, demonstrating that deep learning models can generalize well with moderate dataset sizes. The dataset size (50 Cirrus and 50 Spectralis volume scans) was selected to balance computational feasibility with sufficient representation of disease variability. Figure 1 (a) shows an example of a real Cirrus B-scan from the Cirrus instrument and (b) a synthetic B-scan mimicking the Spectralis instrument.

3D Deep Learning Classifier

We used a CNN-based 3D ResNet model for 3D glaucoma OCT image classification. The structure of the network is detailed in a previous publication.¹⁵ After data augmentation (B-scans rotation and shifting), the 3D OCT images were processed through the model. Specifically, the 3D OCT images pass through a sequence of layers: convolution, batch normalization (BN), and rectified linear unit (ReLU), twice. This sequence, termed a "Conv3D Block," is followed by five stacked residual blocks to extract feature representations from the 3D OCT scans. Each residual block includes a max-pooling layer to down-sample the input feature matrix and prevent overfitting and comprises three residual units. Each residual unit contains two 3D convolution layers with $3 \times 3 \times 3$ filters, with BN and ReLU layers between the filters. A skip connection adds the input directly to the final ReLU layer after the Convolution and BN layers. After feature learning,

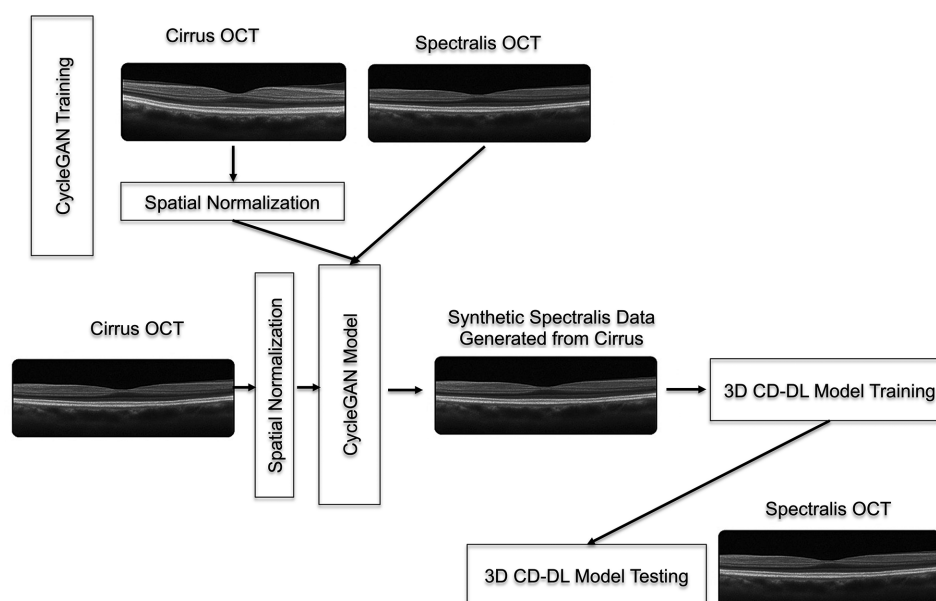


Figure 2. Overview of the 3D CD-DL model design. Cirrus OCT volumes were preprocessed and translated into Spectralis-like images using a CycleGAN trained on unpaired Cirrus and Spectralis scans. The resulting synthetic Spectralis-like B-scans were used to train a 3D convolutional neural network (3D CD-DL model), which was then evaluated on real Spectralis images for glaucoma classification.

all feature matrices are flattened via a fully connected layer.

During optimization, we used the Adam method and a softmax classifier based on cross-entropy loss at the output layer. The model uses ReLU as its nonlinear activation function, effectively addressing the gradient reduction or vanishing problem.

Training and Evaluation

In this study, three models were developed:

1. 3D DL model: trained and tested solely on real Spectralis data as a baseline without cross-domain adaptation
2. 3D CD-DL model: a cross-domain model that first uses CycleGAN to transform Cirrus images into Spectralis-like images. The model is then trained on the synthetic Spectralis data and tested on real Spectralis images to enable effective cross-domain adaptation. For simplicity, we refer to the combined CycleGAN transformation and 3D classifier as the '3D CD-DL model', although the domain adaptation is performed by the CycleGAN, and the classifier itself is trained and evaluated in the target (Spectralis) domain using synthetic data. [Figure 2](#) shows an overview of the 3D CD-DL model design.
3. Cross-device baseline: trained on real Cirrus images and tested directly on real Spectralis

images without any domain adaptation, to assess performance under domain shift

All models used a 3D ResNet framework and were evaluated using fivefold cross-validation only; no independent external test set was used. At each fold, approximately 100–101 glaucoma eyes and 23–24 healthy eyes were used for testing, with the remaining data used for training. For each model, we used four subsets to train the model and used the fifth subset to assess model performance. This sequence was repeated five times, with each subset serving as the test set one time so that each tested patient and eye were never part of its own training set and was tested only once. An augmentation procedure in the form of horizontal mirroring like that used by Christopher et al. was applied to the minority class to balance the data.¹⁶

Statistical Analyses

Descriptive statistics included the calculation of the mean and standard deviation for normally distributed variables and median, first quartile, and third quartile values for non-normally distributed variables. Student's *t*-tests or Mann-Whitney tests were used to evaluate the statistical significance of differences in demographic and clinical parameters between glaucoma patients and healthy individuals.

AUROC were used to evaluate the diagnostic accuracies of the different models investigated in the

study. An AUROC regression model with maximum likelihood estimator was applied to adjust for potentially confounding effects of age, image quality, and axial length.¹⁷

To assess agreement between the 3D DL and 3D CD-DL models, kappa agreement was estimated and interpreted according to Landis.¹⁸ Statistical analyses were performed using Stata 14.2 (StataCorp LLC, College Station, TX, USA). P values ≤ 0.05 were considered statistically significant.

Results

This cross-sectional study included 502 POAG eyes from 295 patients and 119 healthy eyes from 63 study participants. Table 1 displays the demographic and clinical characteristics of the participants and eyes.

Mean (95% CI) participant age at SD-OCT imaging date was 67.3 (95% CI, 65.2–69.3) years for patients with glaucoma and 60.5 (95% CI, 58.2–63.4) years for the healthy individuals. POAG eyes had worse VF MD (−2.64 [95% CI, −3.66 to −1.93]) than healthy eyes (0.23 [95% CI, −0.38 to 0.87]) ($P < 0.001$). Spherical equivalent was slightly more myopic in the healthy group (−1.28 D; 95% CI, −1.91 to −0.18) compared to the POAG group (−1.47 D; 95% CI, −1.99 to −1.15), although this difference was not statistically significant ($P = 0.21$). There was no significant difference in spherical equivalent ($P = 0.21$), axial length ($P = 0.32$), or IOP ($P = 0.43$) between groups.

AUROC and sensitivities at 80% and 95% specificity for all four models—3D CD-DL, 3D DL, GCIPL thickness, and the cross-device baseline—are provided in Table 2. The 3D CD-DL model, trained on synthetic Spectralis-like B-scans generated from Cirrus images, achieved an adjusted AUROC of 0.91 (95% CI,

Table 1. Study Participant and Eye Characteristics by Diagnosis

	Diagnosis		P Value
	Healthy ($N = 63$ Persons, 119 Eyes)	Glaucoma ($N = 295$ Persons, 502 Eyes)	
Age (years)	60.5 (58.2, 63.4)	67.3 (65.2, 69.3)	<0.001
Sex (% Female)	65.1%	50.1%	0.02
Race (%)			
Non-White	52.3%	62.7%	0.002
White	47.7%	47.3%	
VF MD (dB)	0.23 (−0.38, 0.87)	−2.64 (−3.66, −1.93)	<0.001
IOP (mm Hg)	15.6 (13.9, 16.4)	14.3 (13.9, 15.4)	0.43
AL (mm)	23.8 (23.5, 25.1)	24.0 (23.8, 24.7)	0.32
Spherical equivalent (D)	−1.28 (−1.91, −0.18)	−1.47 (−1.91, −1.08)	0.21

AL, axial length.

Mean values and 95% confidence intervals are shown for continuous variables. Statistical significance of differences in continuous and categorical variables are determined by two-sample t -tests and Fisher's exact tests for patient level variables (respectively) and linear mixed effects models for eye level variables.

Table 2. Estimated Diagnostic Performance of All Models Evaluated in the Study, Including AUROC, Accuracy, Precision, and Specificity at Fixed Sensitivity Thresholds

	AUROC (95% CI)	Accuracy (95% CI)	Precision (95% CI)	Specificity at 95% Sens (95% CI)	Specificity at 80% Sens (95% CI)	Adjusted AUROC P Value Compared to 3D DL
3D DL	0.92 (0.85, 0.95)	90.7% (87.2, 94.2)	87.0% (82.5, 91.5)	84.3% (81.2, 87.4)	74.2% (70.1, 78.1)	0.45
3D CD-DL	0.91 (0.84, 0.95)	90.4% (86.9, 93.9)	86.7% (82.2, 91.2)	83.9% (80.3, 86.9)	72.9% (69.3, 76.8)	
Cross-device baseline	0.82 (0.77, 0.84)	81.8% (74.3, 84.3)	74.6% (70.1, 78.2)	68.5% (62.1, 71.4)	55.2% (47.0, 59.5)	<0.001
GCIPL global thickness	0.83 (0.78, 0.85)	82.1% (78.6, 85.6)	75.5% (71.0, 80.0)	69.2% (64.2, 72.3)	56.3% (47.1, 58.3)	<0.001

All eyes ($n = 621$) included.

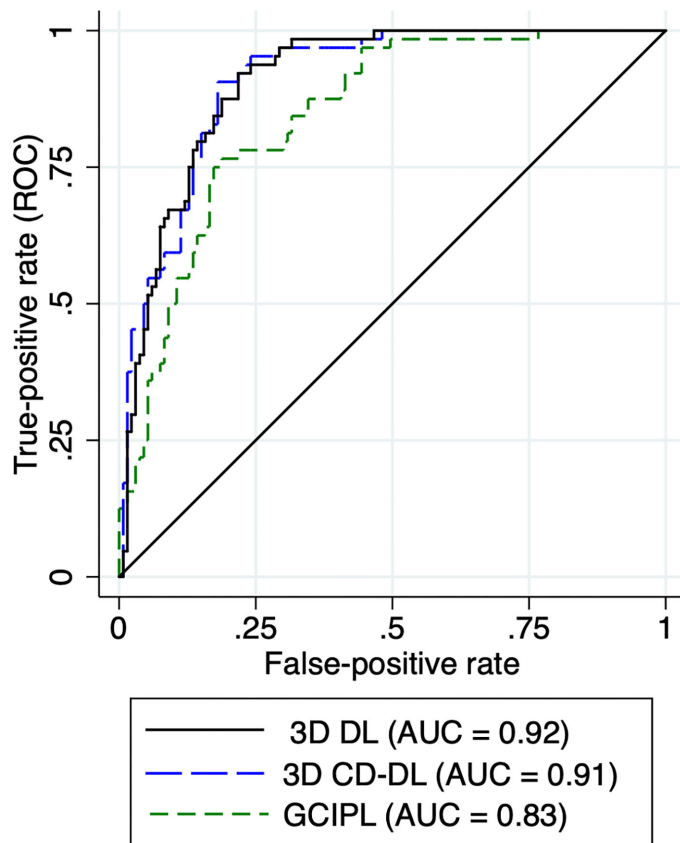


Figure 3. Receiver operating characteristic curves to discriminate glaucomatous from healthy eyes for the three classifiers used: the standard macular GCIPL thickness, 3D DL, and 3D CD-DL.

0.84–0.95). This performance was statistically equivalent to the device-specific 3D DL model (AUROC = 0.92; 95% CI, 0.85–0.95; $P = 0.45$), and both significantly outperformed GCIPL thickness (AUROC = 0.83; 95% CI, 0.78–0.85; $P \leq 0.001$). The cross-device baseline, trained on Cirrus and tested directly on Spectralis without adaptation, had the lowest AUROC at 0.82 (95% CI, 0.77–0.84; $P < 0.001$).

In addition to AUROC, accuracy and precision were assessed in Table 2. The 3D CD-DL model achieved an accuracy of 90.4% (95% CI, 86.9–93.9) and a precision of 86.7% (95% CI, 82.2–91.2), closely matching the 3D DL model (accuracy = 90.7% [95% CI, 87.2–94.2]; precision = 87.0% [95% CI, 82.5–91.5]). GCIPL thick-

ness showed lower performance (accuracy = 82.1% [95% CI, 78.6–85.6]; precision = 75.5% [95% CI, 71.0–80.0]), whereas the cross-device baseline performed worst (accuracy = 81.8% [95% CI, 74.3–84.3]; precision = 74.6% [95% CI, 70.1–78.2]).

Figure 3 shows the AUROC plot for the three classifiers: 3D DL, 3D CD-DL and the GCIPL thickness. Performance results for early glaucoma eyes are summarized in Table 3. Specifically, in eyes with early glaucoma ($MD \geq -3$ dB, case: control ratio $\sim 3:1$), which included 351 POAG eyes from 217 patients and all 119 healthy eyes from 63 study participants, the 3D DL model demonstrated significantly higher diagnostic accuracy (AUROC of 0.90 [95% CI, 0.84–0.94]) compared to global GCIPL thickness (AUROC of 0.80 [95% CI, 0.76–0.82]) ($P \leq 0.001$). However, this accuracy was similar to that of the 3D CD-DL model, which had an AUROC of 0.90 (95% CI, 0.84–0.93) ($P = 0.51$).

We also evaluated model performance in eyes with more advanced glaucoma ($MD < -3$ dB; case: control ratio $\sim 1.2:1$ $N = 151$), which included 151 POAG eyes from 78 patients and all 119 healthy eyes from 63 study participants. In this subgroup, the 3D CD-DL and 3D DL models continued to perform well, achieving AUROCs of 0.95 and 0.96, respectively ($P = 0.51$). GCIPL thickness was less accurate, with an AUROC of 0.91 ($P < 0.001$), and the cross-device baseline model again showed the lowest performance with an AUROC of 0.89 ($P < 0.001$). There was substantial agreement between models in classifying healthy versus glaucoma, with an exact agreement of 95% (95% CI, 93%–97%) and a kappa coefficient of 0.78 (95% CI, 0.75–0.82).

Discussion

The current study compared the diagnostic accuracy of two 3D DL classifiers—one trained on Spectralis images and a cross-domain DL model trained on synthetic data generated from Cirrus images—against standard macular GCIPL thickness measurements. Results demonstrate that the

Table 3. Estimated AUROC and Sensitivities at Fixed Specificities for All Models

	AUROC (95% CI)	80% Specificity	95% Specificity	Adjusted AUROC P Value Compared to 3D DL
3D DL	0.90 (0.84, 0.94)	81.4% (75.4, 86.2)	70.4% (66.8, 74.2)	
3D CD-DL	0.90 (0.83, 0.93)	80.3% (74.3, 83.8)	69.2% (65.2, 73.9)	0.51
GCIPL global thickness	0.80 (0.76, 0.82)	61.5% (59.4, 66.7)	53.2% (45.4, 55.1)	<0.001

Early glaucoma eyes ($MD \geq -3$) eyes ($n = 470$) included.

device-specific 3D DL model achieved significantly higher accuracy than GCIPL thickness with an adjusted AUROC ≥ 0.91 compared to 0.83 for GCIPL. The cross-domain 3D CD-DL model, trained using synthetic data generated with CycleGAN, performed similarly to the device-specific 3D DL model, with an AUROC of 0.91. This highlights the potential for cross-domain learning to generalize across different imaging platforms, which could reduce the need for device-specific models in clinical practice.

Importantly, the agreement between the 3D DL and 3D CD-DL models was substantial, with approximately 95% agreement in classifying healthy versus glaucomatous eyes, and a kappa value of 0.78. This high level of agreement suggests that the cross-domain model performs, as well as the device-specific model in classifying glaucoma, further supporting the feasibility of using such models across diverse clinical settings.

In early glaucoma ($MD \geq -3$ dB), the 3D DL model significantly outperformed GCIPL thickness measurements, with an AUROC of 0.90 compared to 0.80 for GCIPL, underscoring the potential for deep learning models to detect early glaucomatous changes more effectively than traditional structural metrics.

Results from the current study confirm results from other studies that highlight the value of DL analyses for glaucoma detection. For example, as previously described, Mehta et al.⁷ developed a multi-modal model that integrated ocular and clinical data to detect glaucoma. Their OCT-based model used a 2D-CNN framework with a Densenet201 backbone, trained on individual B-scans from OCT images. This model achieved AUROC of 0.95 by aggregating predictions from each B-scan. In addition, Maetschke et al.⁹ explored a 3D-CNN approach for glaucoma detection. Their model classified healthy and glaucomatous eyes directly from unsegmented OCT volumes of the optic nerve head, achieving an AUROC of 0.94. Chiang et al.¹⁹ applied a 3D DL model to automatically segmented wide-field OCT scans to compare diagnostic value across ONH, macula, and combined regions. Although that study focused on segmented tissue structures rather than raw OCT data, it further illustrates the growing application of 3D deep learning to structural OCT for glaucoma-related tasks.

To the best of our knowledge, this is the first study to investigate a cross-domain deep learning model for glaucoma detection. OCT-based cross-domain applications have been explored previously, particularly in segmentation tasks. In particular, Wu et al.¹⁰ developed an unsupervised cross-domain segmentation model, GANSeg, to segment intraretinal fluid and retinal layers from OCT images across different manufacturers, including Heidelberg Engineering (Heidelberg,

Germany), Topcon Medical Systems Inc. (Oakland, NJ, USA), and Zeiss (Oberkochen, Germany) devices. This approach used GANs to generalize a model trained on Heidelberg Spectralis data for segmentation of B-scans from Topcon 1000, without requiring any labeled Topcon data. GANSeg achieved performance comparable to human graders and effectively generalized to unseen devices, such as the Zeiss Plex Elite 9000 and Topcon Maestro2, underscoring the potential of cross-domain learning in expanding the utility of deep learning algorithms across imaging platforms.

To address the limitation of needing large, labeled training datasets, transfer learning has been utilized in several studies.^{20–22} Although transfer learning can reduce the amount of new labeled data required, it still demands a significant amount of manual labeling and retraining when applied to devices with different imaging resolutions or newer technology. This remains a challenge in maintaining scalability as devices evolve over time. In contrast, our approach reduces the need for extensive labeled data from each new device by leveraging a CycleGAN to generate synthetic data resembling the target domain. Although a modest labeled dataset from the source device (e.g., Cirrus) is still required to train the CycleGAN, this method minimizes the need for large amount of manually labeled data from the target device (e.g., Spectralis), improving scalability across platforms. This makes it a more feasible and efficient solution for clinical applications.

There are several possible limitations to the current study. First, although fivefold cross-validation with independent test and training sets was used to control for overfitting of models, the 3D DL CNN models were not tested on external test datasets, so accuracy may be reduced in other populations. Second, the relatively small sample size (502 glaucoma eyes, 119 healthy eyes) may limit the generalizability of our findings, although the sample size was large enough to detect significant differences among analysis strategies. Third, although the model was validated on the Cirrus OCT device, it is important to evaluate its performance across a wider range of imaging platforms, including newer instruments, to ensure broad applicability. Last, although the model reduces the need for manual labeling, a modest training set is still required for each new instrument. This need could be a challenge when adapting the model to future devices.

In addition, the study dataset includes a higher proportion of glaucoma eyes than would be found in typical screening populations, where disease prevalence is $<5\%$. This imbalance may inflate performance metrics and limit the generalizability of the results to population-based screening settings.

Another limitation is the need for alignment preprocessing to match Cirrus scans to Spectralis. This included cropping a 4.5×4.5 mm fovea-centered region from each Cirrus volume and selecting 73 evenly spaced B-scans. Although necessary for consistent CycleGAN training, this step adds complexity and may limit scalability. Although horizontal mirroring was used for data augmentation, which may help with minor misalignments, CycleGAN performance could still be affected by common acquisition artifacts such as decentration or scan tilt.

Finally, in our study is the definition of early glaucoma as $MD \geq -3$ dB, whereas most studies classify early glaucoma as $MD > -6$ dB. This threshold was selected because our dataset was skewed toward very early-stage glaucoma. Although this narrower definition may limit direct comparisons with other studies, it allowed for a more precise evaluation of model performance in detecting subtle glaucomatous changes. Future studies should investigate the model's generalizability across a wider range of disease severity, including cases with MD between -3 dB and -6 dB or more advanced stages.

In conclusion, this study extends previous research on cross-domain learning with synthetic images in OCT imaging by applying it to glaucoma classification, offering a promising solution to the challenges of device-specific deep learning models. By using unpaired synthetic data generated from Cirrus images, the 3D CD-DL model achieved diagnostic performance on par with device-specific models, supporting its potential use in broader clinical settings. Further validation is needed to assess the model's generalizability across other OCT platforms and to track its performance over time in longitudinal studies.

Acknowledgments

Supported by the National Institutes of Health grants R01EY011008, R01EY019869, U10EY014267, R01EY027510, R01EY026574, P30EY022589, R21EY031125, R01EY034146, R41EY034424, R01EY023704, R01EY029058, R01EY022039, R21EY027945, K99/R00EY030942; BrightFocus Foundation by the donors of the National Glaucoma Research Program [grant number 2017122]; an unrestricted grant from Research to Prevent Blindness (New York, NY); and The Glaucoma Foundation grant (no grant number).

Disclosure: **A. Belghith**, None; **C. Bowd**, None; **R.N. Weinreb**, AbbVie (C), Alcon (C), Balance (C), Eyenovia (C), Toku (C), Topcon (C, R), Center Vue

(R), Konan Medical (R), Optovue (R), Vasoptic (R), Visionix (R), National Eye Institute (R), National Institute of Minority Health (R); **J. Jalili**, None; **M. Christopher**, AISight Health (I, O, P); **L.M. Zangwill**, AbbVie (C), Topcon (C, S), NEI (R), NIH (R), The Glaucoma Foundation (R), Heidelberg Engineering (R, S), DRCR Retina Network/JAEB Center for Health Research (R), The Krupp Foundation (R); Optomed (S), ICare (S), Carl Zeiss Meditec (S), Optovue/Visionix (S), AISight Health (I, O, P)

References

1. Jayaram H, Kolko M, Friedman DS, Gazzard G. Glaucoma: now and beyond. *Lancet*. 2023;402(10414):1788–1801.
2. Zhang X, Jiang J, Kong K, et al. Optic neuropathy in high myopia: glaucoma or high myopia or both? *Prog Retin Eye Res*. 2024;99:101246.
3. Shah M. Micro-invasive glaucoma surgery—an interventional glaucoma revolution. *Eye Vision*. 2019;6:1–5.
4. Tham Y-C, Li X, Wong TY, Quigley HA, Aung T, Cheng C-Y. Global prevalence of glaucoma and projections of glaucoma burden through 2040: a systematic review and meta-analysis. *Ophthalmology*. 2014;121:2081–2090.
5. Weinreb RN, Aung T, Medeiros FA. The pathophysiology and treatment of glaucoma: a review. *JAMA*. 2014;311:1901–1911.
6. Christopher M, Belghith A, Weinreb RN, et al. Retinal nerve fiber layer features identified by unsupervised machine learning on optical coherence tomography scans predict glaucoma progression. *Invest Ophthalmol Vis Sci*. 2018;59:2748–2756.
7. Mehta P, Petersen CA, Wen JC, et al. Automated detection of glaucoma with interpretable machine learning using clinical data and multimodal retinal images. *Am J Ophthalmol*. 2021;231:154–169.
8. Bowd C, Belghith A, Rezapour J, et al. Multimodal deep learning classifier for primary open angle glaucoma diagnosis using wide-field optic nerve head cube scans in eyes with and without high myopia. *J Glaucoma*. 2023;32:841–847.
9. Maetschke S, Antony B, Ishikawa H, Wollstein G, Schuman J, Garnavi R. A feature agnostic approach for glaucoma detection in OCT volumes. *PLoS One*. 2019;14(7):e0219126.
10. Wu Y, Olvera-Barrios A, Yanagihara R, et al. Training deep learning models to work on multiple devices by cross-domain learning with no addi-

- tional annotations. *Ophthalmology*. 2023;130:213–222.
11. Sample PA, Girkin CA, Zangwill LM, et al. The African Descent and Glaucoma Evaluation Study (ADAGES): design and baseline data. *Arch Ophthalmol*. 2009;127:1136–1145.
 12. Bowd C, Belghith A, Rezapour J, et al. Diagnostic accuracy of macular thickness map and texture en face images for detecting glaucoma in eyes with axial high myopia. *Am J Ophthalmol*. 2022;242:26–35.
 13. Medeiros FA, Sample PA, Zangwill LM, Liebmann JM, Girkin CA, Weinreb RN. A statistical approach to the evaluation of covariate effects on the receiver operating characteristic curves of diagnostic tests in glaucoma. *Invest Ophthalmol Vis Sci*. 2006;47:2520–2527.
 14. Zhu J-Y, Park T, Isola P, Efros AA. Unpaired image-to-image translation using cycle-consistent adversarial networks. In: *Proceedings of the IEEE international conference on computer vision*. Piscataway, NJ: IEEE. 2017:2223–2232.
 15. Yang M, Huang X, Huang L, Cai G. Diagnosis of Parkinson's disease based on 3D ResNet: the frontal lobe is crucial. *Biomed Signal Proc Control*. 2023;85:104904.
 16. Christopher M, Bowd C, Belghith A, et al. Deep learning approaches predict glaucomatous visual field damage from OCT optic nerve head en face images and retinal nerve fiber layer thickness maps. *Ophthalmology*. 2020;127:346–356.
 17. Pepe MS. A regression modelling framework for receiver operating characteristic curves in medical diagnostic testing. *Biometrika*. 1997;84:595–608.
 18. Landis J. The measurement of observer agreement for categorical data. *Biometrics*. 1977;33:159–174.
 19. Chiang CY, Braeu FA, Chuangsuwanich T, et al. Are macula or optic nerve head structures better at diagnosing glaucoma? An answer using artificial intelligence and wide-field optical coherence tomography. *Transl Vis Sci Technol*. 2024;13(1):5.
 20. Christopher M, Belghith A, Bowd C, et al. Performance of deep learning architectures and transfer learning for detecting glaucomatous optic neuropathy in fundus photographs. *Sci Rep*. 2018;8(1):16685.
 21. Asaoka R, Murata H, Hirasawa K, et al. Using deep learning and transfer learning to accurately diagnose early-onset glaucoma from macular optical coherence tomography images. *Am J Ophthalmol*. 2019;198:136–145.
 22. Hemelings R, Elen B, Barbosa-Breda J, et al. Accurate prediction of glaucoma from colour fundus images with a convolutional neural network that relies on active and transfer learning. *Acta Ophthalmol*. 2020;98(1):e94–e100.